

Rigorous Analysis of the Triangle Inequality for the NB Dissimilarity Measure

Mathematical Proof and Disproof Report

May 21, 2026

Abstract

This report provides a complete, rigorous mathematical analysis of the triangle inequality for the proposed *Normalized Bounded (NB) dissimilarity measure* defined in Equation (22) of the paper. We prove analytically that the triangle inequality holds strictly in the one-dimensional case ($p = 1$). Conversely, we disprove the validity of the triangle inequality for all higher-dimensional spaces ($p \geq 2$) by presenting a concrete, integer-based 2D counterexample. Finally, we provide the geometric intuition explaining why the triangle inequality is violated in higher dimensions.

1 Introduction and Definition

The p -dimensional NB dissimilarity measure for two vectors $x, y \in \mathbb{R}^p$ (not simultaneously zero) is defined as:

$$d(x, y) = \frac{1}{2} \frac{\sum_{i=1}^p |x_i - y_i| (|x_i| + |y_i|)}{\sum_{i=1}^p (x_i^2 + y_i^2)} \quad (1)$$

If $x = y = 0$, the measure is defined as $d(0, 0) = 0$.

2 Proof of the One-Dimensional Case ($p = 1$)

For $p = 1$, the function simplifies to:

$$d(x, y) = \frac{1}{2} \frac{|x - y| (|x| + |y|)}{x^2 + y^2} \quad (2)$$

Due to the scale-invariance of the function—namely $d(kx, ky) = d(x, y)$ for all $k \neq 0$ —and its negation symmetry $d(-x, -y) = d(x, y)$, we can assume without loss of generality that at least two of the three variables $\{x, y, z\}$ are non-negative. This leads to two primary sign configurations to establish the triangle inequality:

$$d(x, z) \leq d(x, y) + d(y, z) \quad (3)$$

2.1 Piecewise Formulation by Relative Signs

For any $u, v \in \mathbb{R}$:

1. **Same Sign** ($uv \geq 0$):

$$d(u, v) = \frac{1}{2} \frac{|u^2 - v^2|}{u^2 + v^2} \quad (4)$$

2. **Opposite Signs** ($uv < 0$): Assuming $u > 0$ and $v < 0$, let $v = -b$ with $b > 0$. Then:

$$d(u, v) = \frac{1}{2} \frac{(u + b)^2}{u^2 + b^2} = \frac{1}{2} + \frac{|uv|}{u^2 + v^2} \quad (5)$$

2.2 Case A: $x, y, z > 0$ (All Positive)

Let $A = x^2, B = y^2, C = z^2$. Since $x, y, z > 0$, we have $A, B, C > 0$. The dissimilarity measures are:

$$d(x, y) = \frac{1}{2} \frac{|A - B|}{A + B}, \quad d(y, z) = \frac{1}{2} \frac{|B - C|}{B + C}, \quad d(x, z) = \frac{1}{2} \frac{|A - C|}{A + C} \quad (6)$$

We wish to prove:

$$\frac{|A - C|}{A + C} \leq \frac{|A - B|}{A + B} + \frac{|B - C|}{B + C} \quad (7)$$

By symmetry, assume without loss of generality that $A \geq C$. We analyze the three orderings of B :

2.2.1 Ordering 1: $A \geq B \geq C$

Here, $|A - C| = A - C$, $|A - B| = A - B$, and $|B - C| = B - C$. The difference between the RHS and the LHS is:

$$\begin{aligned} \text{Diff} &= \left(\frac{A - B}{A + B} + \frac{B - C}{B + C} \right) - \frac{A - C}{A + C} \\ &= \frac{(A - B)(B + C) + (B - C)(A + B)}{(A + B)(B + C)} - \frac{A - C}{A + C} \\ &= \frac{2B(A - C)}{(A + B)(B + C)} - \frac{A - C}{A + C} \\ &= (A - C) \left[\frac{2B(A + C) - (A + B)(B + C)}{(A + B)(B + C)(A + C)} \right] \end{aligned}$$

Expanding and factoring the numerator inside the brackets:

$$2AB + 2BC - (AB + AC + B^2 + BC) = AB + BC - AC - B^2 = (A - B)(B - C)$$

Thus, we obtain:

$$\text{Diff} = \frac{(A - C)(A - B)(B - C)}{(A + B)(B + C)(A + C)} \quad (8)$$

Since $A \geq B \geq C > 0$, all factors in the numerator are non-negative, and the denominator is strictly positive. Therefore, $\text{Diff} \geq 0$, and the inequality holds.

2.2.2 Ordering 2: $B \geq A \geq C$

Let $f(u) = \frac{u - C}{u + C} = 1 - \frac{2C}{u + C}$ for $u > 0$. Since $f'(u) = \frac{2C}{(u + C)^2} > 0$, $f(u)$ is strictly increasing. Because $B \geq A \geq C > 0$, we have $f(B) \geq f(A)$, meaning:

$$\frac{A - C}{A + C} \leq \frac{B - C}{B + C} \leq \frac{B - A}{B + A} + \frac{B - C}{B + C} \quad (9)$$

which satisfies the inequality.

2.2.3 Ordering 3: $A \geq C \geq B$

Let $g(u) = \frac{A - u}{A + u} = \frac{2A}{A + u} - 1$ for $u > 0$. Since $g'(u) = -\frac{2A}{(A + u)^2} < 0$, $g(u)$ is strictly decreasing. Because $A \geq C \geq B > 0$, we have $g(C) \leq g(B)$, meaning:

$$\frac{A - C}{A + C} \leq \frac{A - B}{A + B} \leq \frac{A - B}{A + B} + \frac{C - B}{C + B} \quad (10)$$

which satisfies the inequality.

2.3 Case B: $x, z > 0$ and $y < 0$ (Opposite Sign in the Middle)

Let $A = x^2, B = y^2, C = z^2$.

- The pairs (x, y) and (y, z) have opposite signs, so $d(x, y) = \frac{1}{2} + \frac{|xy|}{A+B} > \frac{1}{2}$ and $d(y, z) = \frac{1}{2} + \frac{|yz|}{B+C} > \frac{1}{2}$.
- The pair (x, z) has the same sign, so $d(x, z) = \frac{1}{2} \frac{|A-C|}{A+C} < \frac{1}{2}$ (since $|A - C| < A + C$).

Therefore, we have:

$$d(x, z) < \frac{1}{2} < 1 < d(x, y) + d(y, z) \quad (11)$$

which holds with a strict margin of more than $\frac{1}{2}$.

2.4 Case C: $x, y > 0$ and $z < 0$ (Opposite Sign at One End)

Let $z = -w$ with $w > 0$. We have $x, y, w > 0$. Let $A = x^2, B = y^2, C = w^2$. The dissimilarity measures are $d(x, y) = \frac{1}{2} \frac{|A-B|}{A+B}$, $d(y, z) = \frac{1}{2} + \frac{yw}{B+C}$, and $d(x, z) = \frac{1}{2} + \frac{xw}{A+C}$. Subtracting $\frac{1}{2}$ from both sides, the inequality simplifies to:

$$\frac{xw}{x^2 + w^2} \leq \frac{1}{2} \frac{|x^2 - y^2|}{x^2 + y^2} + \frac{yw}{y^2 + w^2} \quad (12)$$

By symmetry, assume $x \geq y > 0$. We want to show:

$$\frac{xw}{x^2 + w^2} - \frac{yw}{y^2 + w^2} \leq \frac{1}{2} \frac{x^2 - y^2}{x^2 + y^2} \quad (13)$$

We analyze two subcases based on $w > 0$:

2.4.1 Subcase 1: $x \geq y \geq w > 0$

Let $g(u) = \frac{uw}{u^2 + w^2}$ for $u > 0$. The derivative is $g'(u) = w \frac{w^2 - u^2}{(u^2 + w^2)^2}$. Since $u \geq w$, we have $g'(u) \leq 0$, meaning $g(u)$ is strictly decreasing. Since $x \geq y \geq w$, we have $g(x) \leq g(y)$, which implies:

$$\frac{xw}{x^2 + w^2} - \frac{yw}{y^2 + w^2} \leq 0 \leq \frac{1}{2} \frac{x^2 - y^2}{x^2 + y^2} \quad (14)$$

which satisfies the inequality.

2.4.2 Subcase 2: $w \geq x \geq y > 0$

By scale-invariance, let $x' = x/y \geq 1$ and $w' = w/y \geq x'$. Let $a = (x')^2 \geq 1$ and $t = (w')^2 \geq a$. The inequality is:

$$\frac{\sqrt{at}}{a+t} - \frac{\sqrt{t}}{1+t} \leq \frac{1}{2} \frac{a-1}{a+1} \quad (15)$$

Let $\sqrt{t} = e^\theta$ and $\sqrt{a} = e^\alpha$ where $\theta \geq \alpha \geq 0$. This can be rewritten using hyperbolic functions:

$$\frac{1}{2} \left[\frac{1}{\cosh(\theta - \alpha)} - \frac{1}{\cosh \theta} \right] \leq \frac{1}{2} \frac{a-1}{a+1} \quad (16)$$

Since $\cosh u$ is strictly increasing for $u \geq 0$, and for $\theta \geq \alpha \geq 0$ we have $\theta \geq \theta - \alpha \geq 0$, we have:

- $\frac{1}{\cosh(\theta - \alpha)} \leq 1$
- $\cosh \theta \geq \cosh \alpha \implies -\frac{1}{\cosh \theta} \leq -\frac{1}{\cosh \alpha}$

Adding these bounds yields:

$$\frac{1}{\cosh(\theta - \alpha)} - \frac{1}{\cosh \theta} \leq 1 - \frac{1}{\cosh \alpha} = \frac{(\sqrt{a} - 1)^2}{a + 1} \quad (17)$$

We now verify if the upper bound satisfies the inequality:

$$\begin{aligned} \frac{1}{2} \frac{(\sqrt{a} - 1)^2}{a + 1} \leq \frac{1}{2} \frac{a - 1}{a + 1} &\iff (\sqrt{a} - 1)^2 \leq (\sqrt{a} - 1)(\sqrt{a} + 1) \\ &\iff \sqrt{a} - 1 \leq \sqrt{a} + 1 \\ &\iff -1 \leq 1 \end{aligned}$$

which is always true.

Thus, the triangle inequality holds rigorously in all cases for $p = 1$.

3 Disproof of the Multi-Dimensional Case ($p \geq 2$)

For dimensions $p \geq 2$, the triangle inequality is **not** preserved.

3.1 2D Integer Counterexample ($p = 2$)

Let us define three vectors $x, y, z \in \mathbb{R}^2$ as follows:

- $x = (-3, -1)$
- $y = (-3, 3)$
- $z = (1, 3)$

3.1.1 Computation of $d(x, z)$:

$$\begin{aligned} |x - z| &= (|-3 - 1|, |-1 - 3|) = (4, 4) \\ |x| + |z| &= (|-3| + |1|, |-1| + |3|) = (4, 4) \\ \text{Numerator} &= \frac{1}{2} \sum_{i=1}^2 |x_i - z_i|(|x_i| + |z_i|) = \frac{1}{2} [4(4) + 4(4)] = 16 \\ \text{Denominator} &= \sum_{i=1}^2 (x_i^2 + z_i^2) = [(-3)^2 + 1^2] + [(-1)^2 + 3^2] = 10 + 10 = 20 \\ d(x, z) &= \frac{16}{20} = 0.80 \end{aligned}$$

3.1.2 Computation of $d(x, y)$:

$$\begin{aligned} |x - y| &= (|-3 - (-3)|, |-1 - 3|) = (0, 4) \\ |x| + |y| &= (|-3| + |-3|, |-1| + |3|) = (6, 4) \\ \text{Numerator} &= \frac{1}{2} \sum_{i=1}^2 |x_i - y_i|(|x_i| + |y_i|) = \frac{1}{2} [0(6) + 4(4)] = 8 \\ \text{Denominator} &= \sum_{i=1}^2 (x_i^2 + y_i^2) = [(-3)^2 + (-3)^2] + [(-1)^2 + 3^2] = 18 + 10 = 28 \\ d(x, y) &= \frac{8}{28} = \frac{2}{7} \approx 0.2857 \end{aligned}$$

3.1.3 Computation of $d(y, z)$:

$$\begin{aligned}
|y - z| &= (|-3 - 1|, |3 - 3|) = (4, 0) \\
|y| + |z| &= (|-3| + |1|, |3| + |3|) = (4, 6) \\
\text{Numerator} &= \frac{1}{2} \sum_{i=1}^2 |y_i - z_i|(|y_i| + |z_i|) = \frac{1}{2}[4(4) + 0(6)] = 8 \\
\text{Denominator} &= \sum_{i=1}^2 (y_i^2 + z_i^2) = [(-3)^2 + 1^2] + [3^2 + 3^2] = 10 + 18 = 28 \\
d(y, z) &= \frac{8}{28} = \frac{2}{7} \approx 0.2857
\end{aligned}$$

3.1.4 Comparison of Dissimilarity Measures:

Sum of sub-dissimilarity measures:

$$d(x, y) + d(y, z) = \frac{2}{7} + \frac{2}{7} = \frac{4}{7} \approx 0.5714 \quad (18)$$

Direct dissimilarity measure:

$$d(x, z) = 0.80 \quad (19)$$

Comparing the two yields:

$$0.80 > 0.5714 \implies d(x, z) > d(x, y) + d(y, z) \quad (20)$$

Thus, the triangle inequality is violated by a margin of ≈ 0.2286 .

4 Geometric Intuition of the Violation

The failure in higher dimensions is caused by the combination of scale-normalization and axis-aligned path partitioning:

1. **Direct Path** ($x \rightarrow z$): Both dimensions change simultaneously, meaning both components are active in the numerator while the total quadratic energy in the denominator is low (20).
2. **Partitioned Path** ($x \rightarrow y \rightarrow z$): The step $x \rightarrow y$ only changes component 2, while component 1 remains constant at -3 . The step $y \rightarrow z$ only changes component 1, while component 2 remains constant at 3.
3. In each sub-step, the constant (“idle”) component contributes heavily to the quadratic energy sum in the denominator ($2 \times (-3)^2 = 18$ for the first step, and $2 \times 3^2 = 18$ for the second step).
4. This “idle energy” inflates the denominators of the sub-dissimilarity measures, artificially shrinking $d(x, y)$ and $d(y, z)$ so that their sum falls below the direct dissimilarity measure.

5 Conclusion

While the NB dissimilarity measure satisfies all other axioms of a metric space (non-negativity, identity of indiscernibles, and symmetry) and holds for the triangle inequality in 1D, it is **not** a valid metric in $\mathbb{R}^p$ for $p \geq 2$.